

1. Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

1). $\frac{2 - x + x^2 + 4x^3}{(x + 4)^3(x^2 + 4)} =$

2). $\frac{x^2 - 2x + 5}{(x^2 + 9)^2 x^2 (x - 1)} =$

3). $\frac{x^3 + 2x - 1}{(x^2 + 1)^2 (x + 2)} =$

2. For each of the following improper integral, one may use Comparison Test to determine if they are convergent or divergent. Circle the correct answers to complete the arguments of using the comparison test.

- 1). The improper integral $\int_1^{\infty} \frac{1}{\sqrt[7]{x^4 - 0.3}} dx$ is (fill in “convergent” or “divergent”) because $\frac{1}{\sqrt[7]{x^4 - 0.3}}$ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^4}$, D). $\frac{1}{x^7}$, E). $\frac{1}{x^{7/4}}$, F). $\frac{1}{x^{4/7}}$

- 2). The improper integral $\int_1^{\infty} \frac{\sin^2 x}{\sqrt[4]{x^5 + 2}} dx$ is (fill in “convergent” or “divergent”) because $\frac{1}{\sqrt[4]{x^5 + 2}}$ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{5/4}}$, D). $\frac{1}{x^{1/4}}$, E). $\frac{1}{x^{4/5}}$, F). $\frac{1}{x^{1/5}}$,

- 3). The improper integral $\int_1^{\infty} \frac{x^3}{\sqrt{x^8 - 0.3}} dx$ is (fill in “convergent” or “divergent”) because $\frac{x^3}{\sqrt{x^8 - 0.3}}$ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x}$, D). $\frac{1}{x^2}$, E). $\frac{1}{\sqrt{x}}$, F). $\frac{1}{x^8}$

3. 1). Simplify the sum $R_n = \sum_{i=1}^n \left(1 + \frac{6i}{n} - \frac{9i^2}{n^2}\right) \frac{3}{n}$. Write your answer as a function of n without any summation signs.

2). Use your result from Part 1), evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(1 + \frac{6i}{n} - \frac{9i^2}{n^2}\right) \frac{3}{n}$.

3). Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(1 + \frac{6i}{n} - \frac{9i^2}{n^2}\right) \frac{3}{n}$ as a definite integral: $\int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{2cm}} dx$.

4. 1). Simplify the sum $R_n = \sum_{i=1}^n \left(\frac{8i}{n} + \frac{64i^3}{n^3}\right) \frac{4}{n}$. Write your answer as a function of n without any summation signs.

2). Use your result from Part 1), evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{8i}{n} + \frac{64i^3}{n^3}\right) \frac{4}{n}$.

3). Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{8i}{n} + \frac{64i^3}{n^3}\right) \frac{4}{n}$ as a definite integral: $\int_{\underline{\quad}}^{\underline{\quad}} \underline{\hspace{2cm}} dx$.

5. Evaluate the following integrals.

1). $\int \frac{\cos(\ln x)}{x} dx$

2). $\int x^4 e^{-3x^5} dx$

3). $\int_1^2 \frac{x^5 \sqrt{x} + 1}{x^4} dx$

4). $\int \frac{5 - 2xe^x}{x} dx$

5). $\int_1^2 \frac{e^{1/x}}{x^2} dx$

6). $\int \frac{\cos x}{1 + \sin^2 x} dx$

7). $\int \frac{\sec^4 x}{\tan^2 x} dx$

8). $\int \arctan x dx$

9). $\int_0^{\pi/6} \sin^3 \theta \cos^2 \theta d\theta$

10). $\int \tan^3 \theta \sec^6 \theta d\theta$

11). $\int x e^{-2x} dx$

12). $\int x^2 e^x dx$

13). $\int \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx$

14). $\int x^{3/2} \ln x dx$

15). $\int_1^e \frac{\ln x}{\sqrt{x}} dx$

16). $\int x \cos(3x) dx$

17). $\int e^{2x} \sin x dx$

18). $\int e^x \cos x dx$

19). $\int \arcsin x \, dx$

20). $\int x \arcsin x \, dx$

21). $\int \cos^2 x \, dx$

22). $\int \sin^4 x \, dx$

23). $\int \sin(8x) \cos(2x) \, dx$

24). $\int \sin(7x) \sin(3x) \, dx$

25). $\int_0^1 \frac{1}{(3x+1)^4} \, dx$

26). $\int \frac{x+3}{(x-1)(x^2+1)} \, dx$

27). $\int \frac{2x+4}{x^2-2x} \, dx$

28). $\int \frac{4x+2}{x^2-2x-8} \, dx$

29). $\int \frac{4x+2}{x(x^2+1)} \, dx$

30). $\int \frac{\sqrt{x^2-9}}{x^2} \, dx$

31). $\int \frac{x^2}{(4+x^2)^{3/2}} \, dx$

32). $\int_0^3 \frac{x^2}{36-x^2} \, dx$

33). $\int \frac{1}{x^2 \sqrt{x^2-9}} \, dx$

34). $\int \frac{1}{x^2 \sqrt{x^2+16}} \, dx$

35). $\int \frac{1}{x^3 \sqrt{x^2-1}} \, dx$

36). $\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16-x^2}} \, dx$

37). $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cot x}{\sin^2 x} \, dx$

38). $\int_0^2 \frac{x^3}{\sqrt{x^2+4}} \, dx$

6. Evaluate the following improper integrals or show that they are divergent.

1). $\int_1^\infty \frac{(\tan^{-1} x)^3}{x^2+1} \, dx$

2). $\int_1^4 \frac{1}{(x-1)^{2/3}} \, dx$

3). $\int_1^\infty \frac{3}{(x+1)^3} \, dx$

4). $\int_{-2}^0 \frac{1}{(x+2)^2} \, dx$

5). $\int_1^\infty \frac{2}{\sqrt{x+3}} \, dx$

6). $\int_1^\infty x e^{-x^2} \, dx$

7). $\int_2^\infty \frac{1}{x(\ln x)^3} \, dx$

8). $\int_1^\infty \frac{\ln x}{x} \, dx$

7. Find the average value of the function $f(x) = \frac{2x}{(1+x^2)^2}$ on $[0, 2]$.

8. Find the average value of the function $f(x) = \cos^5 x \sin^3 x$ on $[0, \pi/2]$.

9. A rod with density $\rho(x) = 6 + \sin x$ lies on the x -axis between $x = 0$ and $x = \pi$.

- 1). Find the mass of the rod. 2). Find the center of the mass of the rod.
10. The linear density in a rod 8 m long is $\frac{12}{\sqrt{x+1}}$ kg/m, where x is measure in meters from one end of the rod.
- 1). Find the mass of the rod. 2). Find the center of mass of the rod.
- 3). Find the average density of the rod.
11. Sketch the region **in the first quadrant** that is bounded by $y = x + 2$, $y = 8 - x^2$ and $x = 0$, and then find the area of this region.
12. Sketch the region bounded by $y = 2 \sin x$, $y = 2 \cos x$, $x = 0$ and $x = \frac{\pi}{2}$, and then find the area of this region.
13. Sketch the region **in the first quadrant** bounded by $y = 5 - x^2$, $y = \frac{1}{4}x^2$, and $x = 0$, and then find the volume of the solid obtained by rotating this region about x -axis.
14. Sketch the region bounded $x = 1 + y^2$, $x = 0$, $y = 1$, and $y = 2$, and then find the volume of the solid obtained by rotating this region about x -axis.
15. Sketch the region bounded by $y = \ln x$, $y = 0$ and $x = 2$, and then find the volume of the solid obtained by rotating this region about y -axis.
16. Sketch the region bounded by $y = \frac{1}{x}$ and $y = \sqrt{x}$ on $x \in [1, 4]$. and then find the volume of the solid obtained by rotating this region about y -axis.
17. For each of the solids of revolution obtained by rotating the given region by the given rotation axis in 1)-6), complete the following Part a), b) and c).
- a). Sketch the graph of given region.
- b). Set up integrals for computing the volume of by using both disk/washer and cylindrical shell method.
- c). Choose one of the integrals you set up in Part b) to compute the volume of the solid of revolution.
- 1). The region is bounded **in the first quadrant** by $y = 4 - x^2$, $x = 0$ and $y = 0$. The solid is obtained by rotating this region about x -axis.
- 2). The region is bounded by $y = 4x$ and $y = \sqrt{x}$. The solid is obtained by rotating this region about x -axis.
- 3). The region is bounded by $y = e^x$, $x = 1$ and $y = 5$. The solid is obtained by rotation this region about y -axis.
- 4). The region is **in the first quadrant** bounded by $x = \sqrt{y}$, $x = 0$, and $y = 4$. The solid is obtained by rotating this region about y -axis.

- 5). The region is bounded by $y = 2x$, $y = x^2$. The solid is obtained by rotating this region about $y = 4$.
- 6). The region is bounded by $y = 3x$, $y = x^2$. The solid is obtained by rotating this region about $x = 3$.

Formula Sheet

$$\sum_{i=1}^n c = cn, \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6}$$

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^4 + 2n^3 + n^2}{4}$$

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \frac{1}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int_1^\infty \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } p > 1 \\ \text{is divergent} & \text{if } 0 < p \leq 1. \end{cases} \quad \int_0^1 \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } 0 < p < 1 \\ \text{is divergent} & \text{if } p \geq 1. \end{cases}$$

$$m = \int_a^b \rho(x) \, dx, \quad \bar{x} = \frac{\int_a^b x \rho(x) \, dx}{\int_a^b \rho(x) \, dx}$$